

How Should We View the Impact of the AI Technological Revolution on the White-Collar Class?

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Of late, news and commentary about software companies promoting AI programming on a massive scale has flooded the internet, with predictions raining down, each more terrifying than the last. In our earlier articles and answers (see: "Under the AI frenzy, will programmers really lose their jobs in droves?" and "Don't be fooled by capital and its hangers-on — the AI bubble is about to burst"), we criticised these "AI apocalyptic" views. Yet although AI cannot replace the vast majority of white-collar workers, and robots are out of the question for blue-collar workers, it is undeniable that AI technology is revolutionising white-collar work. This technological revolution will have a major impact on the class structure of all industrialised countries. Hence it is necessary to discuss this question in depth.

I. The AI Revolution Will Further Proletarianise the White-Collar "Middle Class"

(1) The White-Collar Class Before the AI Revolution

About a dozen years ago, a debate on the "mental proletarians" (naoli wuchanzhe) broke out in the Chinese left-wing online circles. One side held that white-collar workers are intellectuals, traditionally classed as "petty bourgeoisie," not proletarians. The other side held that white-collar workers own no means of production and are employed by others — they are "mental proletarians," part of the proletariat. The debate reached no consensus, but it did throw the question out there and popularise the concept of the mental proletarian.

As the scale and complexity of production grew, informational and intellectual labour became part of the labour process. The bourgeoisie employs mental labourers to perform these "immaterial" activities. Formally, these labourers are employed by capital, own no means of production, and certainly count as proletarians. But because their labour is the decisive factor in the whole production process, they enjoy considerable autonomy in production — which is why, compared with the workers of large machine industry and the assembly line, they still have many characteristics of pre-capitalist labourers. This is why many people regard white-collar workers as intellectuals, as petty bourgeois.

However, after information technology developed, in many industries the machine has in fact become the core of the labour process. It is not the robot assisting the human, but the human assisting the machine to complete the work. Whether in banking, warehouse management, ticketing, or the internal bookkeeping of enterprises, software systems have since the 1980s gradually taken over these tasks. The repetitive, large-scale mental labour of information processing, calculation, storage, and transmission has been automated. The white-collar worker's job is mainly to provide inputs to these systems and to use their outputs, interacting with the outside world. Imagine how many white-collar workers it would take to sustain such complex production and exchange without computer networks and software systems. For many white-collar workers, then, they have become the "appendage" of the computer system. For these white-collar workers, wages are often low and replaceability high; calling them "mental proletarians" is no exaggeration.

Although the spread of information technology brought proletarianisation to white-collar workers in many industries, the work of building the "machines" — the computer systems including all applications — was still done by hand, especially the most basic code writing. This heavy mental labour still required white-collar workers to tap out line after line of code. Economic globalisation brought a sharp expansion of the scale of production, so demand for software exploded across all industries, and society as a whole has long had an enormous demand for programmers. There are now about 30 million programmers in the world. These programmers are the machine-building workers of the information-labour industry. In software companies, although various specialised positions have appeared — development (frontend, backend), testing, operations, presales — on the whole the work is still hand labour, basically similar to the manufacturing-manufactory stage Marx describes in *Capital*. Some small companies have not even reached the manufactory stage, only the level of "simple cooperation." Hence the software engineer class's dependence on capital is formal, not substantial. Though formally proletarians, in class consciousness and other respects they bear many traits of the "small peasant" or "petty bourgeois" of pre-capitalist society; the internet's nickname for them, "code farmers" (ma nong), is very apt.

Moreover, because computer systems cannot process natural language and have difficulty processing image data, in other industries that widely use software for "heavy mental" labour, many tasks cannot be done by machine systems (software). Between different systems of the same company, or between the systems of different companies, if the interfaces have not been aligned in advance, human beings are basically needed to do the splicing work. Hence these machines are still mostly independent machines, not linked into a complete machine system. The white-collar class's transition from formal to substantial dependence on capital is therefore not yet complete.

(2) The White-Collar Class After the AI Revolution

What the AI technological revolution accomplishes is precisely this: it introduces a general-purpose machine system into informational labour (or intellectual labour), transforming the old way of working so that white-collar workers become appendages of the machine system — completing the transition from formal to substantial dependence.

In the past thirty or forty years, with the development of information technology and the global spread of the internet, most of humanity's knowledge has been digitised and stored as data in interconnected computer systems. Not only has the stock of knowledge been moved from offline to online; the production of new knowledge is also stored directly in digital form on the network. Forums and websites let human knowledge accumulate constantly in the "public space" of the internet, and the open-source software movement broke knowledge free of the limits of private property, making it the common wealth of all humanity.

But precisely this public wealth that people have produced — knowledge, code, e-books, papers — has become the key resource by which machine systems replace white-collar labour. A large language model, in essence, uses the common knowledge of all humanity to train a deep neural network of a certain special structure, compressing vast quantities of knowledge into the parameters of the system, so that the system can produce knowledge in response to the input context (the prompt). We will not go into technical details here; the practical situation is that this system, with the help of suitable prompts, can autonomously produce knowledge within a certain range — writing code, writing articles, composing songs, writing copy, organising charts, and so on. If prompts are organised into a certain workflow through RAG and the like, with calls to various external tools, it becomes the so-called AI Agent system. Such a system can accomplish larger tasks to a certain degree — developing a management system, developing a website, or writing a review paper.

Although, as we noted in our earlier article (Much of the Imagination About AI and Robots Is Unrealistic), the scaling law of current large language models is basically finished and the models' hallucinations cannot be eliminated^[1], these companies are all scrambling to make the AI transition — from AI assisting humans to humans assisting AI, reorganising production around AI systems. Of course, employees have no motivation to optimise themselves out of existence. So what to do? First lay people off, forcing the remaining workforce to actively use AI to complete their work. If it ultimately proves that AI's capabilities are limited, the enterprise can rehire at any time. But if the AI transition proves possible, capital will squeeze out the water step by step, until nothing more can be squeezed. Only then will the target form of the AI transition appear, and the ratio of human to machine labour be fixed.

What does all this mean for the white-collar class?

First, for the kind of white-collar work that can be completed in the digital world and whose results can be self-verified before delivery, AI Agents will bring enormous change. The first in line are

programmers — especially those developing on general-purpose hardware, whose work is done digitally within computer systems, and whose results can be verified through compilation, testing, and the like, ensuring correctness (to a certain degree) of syntax and function. The AI Agent can iterate on the feedback of verification and complete the work autonomously. In these labour processes, the "heavy mental" labour — the actual writing of code — will ultimately be done by machines; human labour will withdraw from this link, becoming the supply of inputs to machine labour (requirements analysis and design), supervising machine labour, inspecting its products, and maintaining the products delivered by the machine systems.

And each of these work links will in turn be gradually replaced by various "AI Agents"; the labour cooperation formerly carried out by workers of different trades will become labour cooperation among multiple AI Agents, assisted by white-collar workers. This is exactly the machine system Marx speaks of:

"A real machine system does not replace the individual machines until the object of labour goes through a connected series of graduated processes, carried out by a chain of mutually complementary machines of different kinds. Here we have again the cooperation by division of labour which is peculiar to manufacture, but this cooperation now presents itself as the combination of machines of a partial nature." (Collected Works of Marx and Engels, Vol. 23, p. 416)

And these AI Agents will very likely all call upon the same large model — the LLM, the large language model, is the prime mover of mental labour, providing general brain power:

"A system of machinery, whether it is based on the mere cooperation of similar machines, as in weaving, or on a combination of different kinds of machines, as in spinning, forms a large automatic machine as soon as it is driven by a self-acting prime mover. ... We have an automatic system of machinery when the working machine performs all the movements necessary for working up the raw material without human help, and requires only attendance from man. The system can, however, be continually improved in its details." (Collected Works of Marx and Engels, Vol. 23, p. 418)

The software systems built by programmers once turned other white-collar workers, to varying degrees, from middle-class into "mental proletarians." This time, the AI transformation is making the programmers themselves race down the road of proletarianisation — even farther than other white-collar groups.

For the other white-collar groups, the impact of AI is likewise broad and profound. The large language model can understand language. For white-collar workers whose job is merely to provide inputs to software systems, a large part of their tasks can be directly replaced. For those whose work involves more complex processing steps, AI systems are not yet enough to replace them directly — mostly they raise efficiency. Because many jobs involve discrete knowledge tied to specific customers

and specific labour processes, knowledge that does not exist on the internet and is constantly changing — if the white-collar worker does not actively codify this knowledge into SKILLS, the AI Agent cannot do the job. Of course, even if the capitalist forces you through various means to solidify your knowledge, much work is in constant flux and requires constantly refreshing the SKILL, so manpower is actually hard to reduce. The likely end result: white-collar workers universally use AI systems to do their work, but they must constantly adjust the input information (the context) to the AI system, changing the semantic background of the model's reasoning. In information production and knowledge production, the machine system centred on AI will in many industries become an indispensable — even central — factor of production.

Marx's description of the factory worker in *Capital* is, it must be said, breathtaking. The white-collar readers of this article might ask themselves: facing this mighty power of the AI system as a scientific achievement, do they not feel their own "insignificance," their "emptiness":

"The specialised skill of the individual machine worker, emptied of meaning, vanishes as an insignificant appendage before science, before the colossal natural forces, before social mass labour, all embodied in the machine system, which together with it constitutes the power of the 'master.'"
(Collected Works of Marx and Engels, Vol. 23, p. 416)

So whether they are the white-collar workers who build the machine systems, the programmers, or the white-collar workers who operate them, the AI system will gradually turn from a "tool" into a "machine": from the worker using the machine to the machine using the worker. The rhythm of the machine will become the rhythm of labour.

As for how many white-collar workers AI will replace, that is hard to estimate; my personal estimate is a reduction of 20%–30% of white-collar positions. But as the scale of production expands and society's needs grow, the overall number of jobs will not fall much — it may even increase. The online talk that most white-collar jobs will disappear within 3–5 years, and that within 10 years humans need not work at all, can be directly judged as nonsense.

II. The Concrete Manifestations of White-Collar "Proletarianisation"

The reconstruction of information-production work by AI systems will greatly affect the lives of white-collar workers. On this point, we can analyse by analogy with the impact of machine-based large-scale production on the workers of manufacture during the First Industrial Revolution.

(1) The Deskilling of White-Collar Workers

We know that the value of labour power is determined by the value of the means of subsistence required for its reproduction. Because white-collar work requires a good deal of knowledge and skill,

white-collar workers are generally university graduates. Apart from deviations caused by supply and demand in the labour market (such as the shortage of blue-collar workers in developed countries), white-collar wages are generally higher than blue-collar, because the cost of university education is included in the white-collar wage.

But the AI Agent system centred on the LLM will markedly lower the knowledge requirements on the worker. In the First Industrial Revolution, machines replaced human muscle power, so capital began to employ women and children on a large scale instead of adult male workers, cutting costs and expanding the labour supply:

"In so far as machinery dispenses with muscular power, it becomes a means of employing workers of slight muscular strength, or whose bodily development is immature, but whose limbs are all the more supple. The first slogan of the capitalist employment of machinery was therefore the labour of women and children! This powerful means of replacing labour and workers thus immediately became a means of subjecting all members of the worker's family, regardless of sex and age, to the direct rule of capital, thereby increasing the number of wage-labourers. The compulsory labour performed for the capitalist not only robbed the children of their playtime, but also robbed the family itself of the free labour time usually required for household purposes." (Collected Works of Marx and Engels, Vol. 23, p. 416)

With an AI system, as long as you can describe the requirement clearly, you can write programs without any programming knowledge. Just as in the First Industrial Revolution, when a few adult male workers led a gang of children and youths, will the capitalists use a few highly skilled old hands (university graduates) to lead a group of high-school graduates to do the work formerly done by several university graduates? Entirely possible. Although the total number of white-collar workers may not decrease, the skill requirements of white-collar work will weaken and wages may fall substantially. If so, society's demand for university graduates will plummet. Hence we also see similar talk, such as the uselessness of university education:

大学文凭成废纸？AI 暴击美国 00 后！他哥大退学成千万富翁，我却还要还学贷



新智元

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【新智元导读】美国 Z 世代⁺表示：因为 ChatGPT，大学文凭已经变成一张废纸！纯属浪费时间浪费金钱。刚刚出炉的调查显示，同意此观点的年轻人已经达到了 49%。而哥大退学网红等 00 后创业明星的故事，更是让美国人对大学教育完全祛魅了。

摊牌了，不装了，美国的 Z 时代已经对上大学彻底不耐烦了。

现在，他们喊出这样的宣言：有了 ChatGPT，大学文凭已经彻底成了一张废纸！纯粹就是浪费时间 and 金钱。

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(2) Longer Working Hours

History has repeatedly proved that tools that raise labour efficiency do not make the labourer more relaxed; they only exhaust him more^[2].

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AI is exhausting workers so much, researchers have dubbed the condition 'AI brain fry'

MAR 13, 2026

Analysis by Allison Morrow

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Marx analysed this phenomenon in depth in Capital:

"If the machine is the most powerful means of raising the productivity of labour, i.e. of shortening the labour time necessary for the production of a commodity, then as the bearer of capital it becomes, first of all, in the industries it directly seizes, the most powerful means of lengthening the working day beyond every natural limit. On the one hand it creates new conditions which allow capital to give free rein to this constant tendency of its, and on the other hand it creates new motives which intensify capital's greed for the labour of others." (Collected Works of Marx and Engels, Vol. 23, pp. 441–442)

As the machine system plays a greater role in the labour process, the workers' resistance is crushed, and the unemployment caused by the adoption of machines leaves workers desperate for any job with no bargaining power. The final result: the working day is greatly extended:

"Hence the capitalist application of machinery, on the one hand, creates new powerful motives for the unlimited lengthening of the working day, and transforms the very mode of labour and the character of the social labour body in such a way that resistance to this tendency is broken; on the other hand, partly by bringing under the sway of capital those strata of workers previously inaccessible to it, and partly by throwing out of work the workers displaced by the machine, it produces a surplus working population, which must obey the laws imposed upon it by capital. Hence there arises a remarkable phenomenon in the history of modern industry: machinery abolishes all moral and natural limits of the working day." (Collected Works of Marx and Engels, Vol. 23, p. 447)

(3) Increased Labour Intensity

Let us first look at how Marx analysed the impact of machine adoption on labour intensity during the First Industrial Revolution:

"The shortening of the working day, which at first created the subjective condition for the condensation of labour, i.e. made it possible for the worker to expend more force in a given time, once legally enforced, placed in the hands of capital the machine as an objective and systematic means of extracting more labour in the same time. This is achieved in two ways: by raising the speed of the machine, and by extending the number of machines under the supervision of the same worker, i.e. the worker's field of labour." (Collected Works of Marx and Engels, Vol. 23, p. 452)

It must be said: what once happened to manual labourers is now being replayed in the field of mental labour. With the spread of AI Agent tools, capitalists demand that workers drive several Agents at once; this multi-task concurrency greatly increases labour intensity and exhausts the worker more^[3].



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AI

Using too many AI tools at once can actually make you less productive and cause 'brain fry,' study finds

By Brent D. Griffiths

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(4) The Intensified Despotism of the Labour Organisation

We know that blue-collar workers are often given lectures before work, and discipline requirements during work are strict. In this respect, white-collar workers have enjoyed a privilege. If the AI system's reconstruction of white-collar work is completed, it is foreseeable that discipline requirements on white-collar work will rise too.

Marx, likewise, made a profound analysis of this phenomenon:

"The technical subordination of the worker to the uniform motion of the instruments of labour, and the peculiar composition of the labour body composed of individuals of both sexes and all ages, created a barrack-like discipline. This discipline developed into a complete factory system, bringing to full development the supervisory labour mentioned earlier, and at the same time the division of workers into labourers and overseers, into ordinary industrial soldiers and industrial non-commissioned officers. 'The bourgeoisie is extremely fond of the separation of powers, and particularly of the representative system; but capital establishes in the factory code its absolute autocracy over its workers by private legislation.' This code is but a capitalist caricature of the social regulation of the labour process, which is necessary for large-scale cooperation and the use of common instruments of labour, especially the machine. The slave-driver's whip has been replaced by the overseer's book of fines. And needless to say, all punishments are simplified into fines and wage deductions, and the legislative wisdom of the factory Lycurguses makes it more advantageous for them to break the law than to keep it." (Collected Works of Marx and Engels, Vol. 23, p. 465)

In the era when heavy mental labour was done mainly by the human brain, labour supervision was difficult, because it is hard to control the worker's actual mental activity. In the labour process, the worker was autonomous. Now this mental activity is accomplished through the interaction and cooperation of humans and machine systems, and the machine performs the general mental labour within it. Capital therefore obtains the means of comprehensive supervision over the labour process. For instance, companies have begun to monitor their employees' token usage^{[4][5]}.

大厂“牛马”，被迫用AI



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这股“AI提效”的风，还是刮到了大厂打工人身上。

最开始，AI还只是少数技术极客和尝鲜者的玩具。有人自掏腰包买会员，有人私下交流提示词，把它当成提高效率的新工具，确实从中尝到了甜头。

嗅一嗅

但现在，情况变了。国内外的互联网大厂已经从“鼓励使用AI”进入了“隐性强制使用AI”的阶段。有人被统计每天消耗了多少Token，有人所在团队把AI使用情况和绩效挂上了钩，有人被要求优先使用公司自研工具，有人则要把自己的工作经验拆成流程、写成Skills，交给AI反复调用。

当“用AI”、“烧Token”逐渐变成一种考核、一套要求，甚至一种新的工作模板，那些被卷进这场智能化浪潮的大厂员工，真实处境到底如何？

这两天，我们和六位来自不同公司、不同岗位的从业者聊了聊。他们的背景涵盖了海外上市公司的CIO、国内头部大厂的高级研发、负责写代码的初级程序员，以及做运营和市场商务的非技术岗。

有人靠着AI实现效率翻倍，将产品需求文档的输出周期从几周压缩到一天，甚至一个人干出了过去一个团队成果；也有人为了应对“智能化产出”的要求，把一份简单的数据看板手动调试了60遍，硬生生把AI用成了需要不断“擦屁股”的初级实习生。

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III. The Resistance of the Mental Proletarians

The capitalists' replacement of white-collar workers by machines will inevitably provoke resistance from the workers. What forms this resistance will take, we can likewise compare with the forms of resistance when handicraft workers were replaced by machines.

(1) Luddism: Resistance Against the Machine Itself

A few days ago we saw a widely circulated post online. What made it go viral was the invention of a new word, "code traitor" (ma jian) — the idea being that the large-model developers have killed everyone.

你们搞大模型的就是码奸，你们已经
害死前端兄弟了，还要害死后端兄
弟，测试兄弟，运维兄弟，害死网安
兄弟，害死ic兄弟，最后害死自己害
死全人类

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The direct antagonism between workers and the machine system is obvious. Although AI Agents are still being deployed and the concrete effect of replacement is not yet clear, many capitalists cannot wait, and have set efficiency-increase targets one after another; it is said many companies have set a 50% efficiency target. If total output stays the same, a 50% efficiency increase means a 50% layoff. Although the layoffs in big companies today are not the result of AI efficiency gains, the bosses of these enterprises are genuinely pushing AI systems into production. Say nothing of whether these targets can be achieved; even if they are never achieved, the process of thrashing about will turn many people's lives upside down — broken families, wives and children scattered. On the one hand a mass of workers unemployed; on the other, the mass of unemployed workers further drives down wages. How can white-collar workers not hate those who develop and introduce AI systems?

On this, Marx's analysis remains penetrating:

"As soon as the instrument of labour appears as a machine, it immediately becomes a competitor of the worker himself. The self-expansion of capital through the machine is proportional to the number of workers whose conditions of existence are destroyed by the machine. The whole system of capitalist production is based on the worker selling his labour power as a commodity. Division of labour renders this labour power one-sided, giving it only the specific skill of manipulating a partial tool. Once the tool is manipulated by the machine, the exchange value of labour power vanishes along with its use value. The worker, like a banknote that has ceased to circulate, cannot be sold. A part of the working class is thus turned by the machine into a surplus population — a population no longer directly needed for the self-expansion of capital — and this population perishes in part in the unequal struggle of old handicraft and manufacture against machine production, and in part floods all the more accessible branches of industry, glutting the labour market and thereby reducing the price of labour power below its value." (Collected Works of Marx and Engels, Vol. 23, p. 471)

If readers know a little history of the machine's large-scale replacement of workers during the Industrial Revolution, they know this is a heart-rending history of blood and tears. The workers destroyed by the machine launched movements to smash the machines.

"To be sure, once these weavers have ended their short lives, the machine brings them only 'a short inconvenience.' Moreover, since the machine constantly occupies new fields of production, its 'short' influence becomes a long one. The capitalist mode of production thus gives labour conditions and labour products a form independent of, and alien to, the worker; and with the development of machinery this form develops into a complete antagonism. Hence, with the appearance of the machine, there occurs for the first time a violent resistance of the workers against the instruments of labour." (Collected Works of Marx and Engels, Vol. 23, pp. 472–473)

Likewise, we see how ecstatic the capitalists are about AI replacing labour. We can see on the internet, on various platforms, how they use AI to beat down the workers.

"But the machine is not only a powerful competitor, always ready to render the wage-worker 'superfluous.' It is also publicly and consciously declared by capital to be a power hostile to the worker, and used as such. The machine becomes the most powerful weapon for suppressing the periodic revolts and strikes of the workers against the despotism of capital. As Gaskell puts it, the steam engine was from the outset an antagonist of 'human power,' enabling capital to crush the workers' rising demands which might have plunged the nascent factory system into crisis. A whole history could be written showing how, since 1830, many inventions have served only as weapons of capital against the revolts of the workers. We need only think of the self-acting mule, which opened a new epoch in the automatic system." (Collected Works of Marx and Engels, Vol. 23, p. 477)

The Luddite movement was, of course, bound to fail. Even if hackers representing the workers' interests blackened all these AI model services, the process could not be stopped. But the workers' resistance is normal; it is the first form of resistance. No one whose arse is not on the wrong side will mock such resistance. As Lenin said:

"There was a time when the workers' hostility to capital manifested itself only as a vague feeling of hatred for their exploiters, as a vague awareness of their oppression and enslavement, and as a desire for revenge upon the capitalists. The struggle then took the form of scattered actions by the workers — wrecking buildings, smashing machines, beating up managers, and so on. This was the initial, nascent form of the working-class movement, and it was necessary at the time, for hatred of the capitalist has at all times and in all places been the first impetus giving rise to the workers' demand for self-defence." (Lenin, "Draft Programme of the Social-Democratic Party and Its Explanation")

If AI Agents prove effective — even replacing only twenty or thirty percent of white-collar workers — in the coming years we will witness the first awakening of class consciousness in the white-collar class.

(2) The Union of White-Collar and Blue-Collar Workers

The so-called "middle class," dominated by white-collar workers, has always been the class base of pro-capitalist forces within the working-class movement. The great differences between white-collar and blue-collar workers in labour conditions, income, and the like made effective unity between these two great groups difficult. Moreover, the existence of white-collar workers led blue-collar families to set the goal of raising a white-collar offspring, giving many workers faith in changing their fate through education. All this kept the capitalist system stable.

The further proletarianisation of the white-collar class brought by AI technology will change this state of affairs. First, the deskilling of white-collar workers, the sharp reduction of well-paid jobs, and job insecurity all strengthen this group's class consciousness. Second, the large-scale disappearance of well-paid white-collar jobs will greatly reduce the possibility of blue-collar workers realising their various "dreams" under the capitalist system. More and more people will recognise that raising one's income through knowledge and skills is becoming harder and harder, that the gulf between capital and labour is widening, and that improving living standards, reducing labour intensity, and shortening labour time can only be achieved through collective struggle, not individual effort. Capitalist society will still offer some well-paid jobs, so some people will keep dreaming; but the gap between dream and reality grows, and more and more people will wake from the dream.

Once, white-collar workers even had the idealistic project of running the open-source software movement within the capitalist system; today, the large-model companies train their models on the free code created by that open-source movement, helping capitalists replace programmers with machines. Capitalism has a tendency to turn everything public, shared, and beautiful into private commodities, into tools of profit-making. From bike-sharing companies appropriating the public space of the streets to large-model companies "freely using" the intellectual products of humanity, this scene repeats itself again and again. Under the hammer blows of capital, white-collar workers will become ever more clear-sighted about their class position. And clear-sightedness is the beginning of all rational action.

On this part, constrained by the environment, we cannot say more. Reviewing the history of two hundred years of the workers' movement will provide us lessons enough. From Bangalore to San Francisco, from Rio de Janeiro to Nairobi, we are all ordinary members of this group. "In the peaceful world, all share one warmth and one coldness" — our fates are bound together.

IV. Summary

To view the AI system's impact on the white-collar group objectively, one needs a twofold understanding:

First, the AI system cannot replace most white-collar workers. Current LLMs cannot achieve AGI. The enhanced AI Agent still requires human labour. Do not be frightened by the capitalists; do not use science-fiction scenarios to fall into catastrophic thinking. It must also be noted that robot technology is far less mature than large language models and still needs a long scientific research process to leave the laboratory; at present it is nowhere near commercial use. See "Much of the Imagination About AI and Robots Is Unrealistic."

Second, one must recognise that the AI system has the potential to comprehensively transform white-collar work, and the end result will very likely be that instead of the white-collar worker using tools to work, the machine (AI) uses the white-collar worker to work. White-collar work will undergo all the changes discussed above, and in living standards and consciousness will gradually approach the blue-collar worker. The changed conditions of production give the two great armies of the working class the real possibility of "joining forces."

Notes

[1] "H-Neurons: On the Existence, Impact, and Origin of Hallucination-Associated Neurons in LLMs" — this paper from Tsinghua University deserves attention. The authors found the artificial neurons that cause hallucinations, but their research also shows that these neurons play a key role in model performance, so hallucinations cannot be solved by simply disabling them. The emergence of H-Neurons is likely caused by the pretraining process of next-token prediction, and is intrinsic to the architecture of large language models.

中文原文：《H-Neurons: On the Existence, Impact, and Origin of Hallucination-Associated Neurons in LLMs》清华大学的这篇论文非常值得关注。这篇论文的 authors，找到了导致幻觉出现的人工神经元，但是他们的研究也表明，这些神经元除了造成幻觉，对模型的性能也起着关键作用，所以无法直接通过禁用这些神经元的方式解决幻觉。H-Neuron 的出现，很可能是下一个词元预测这种预训练过程导致的，是内在于大语言模型架构的。

<https://arxiv.org/abs/2512.01797>

[2] "AI is exhausting workers so much, researchers have dubbed the condition 'AI brain fry'" (CNN)

中文原文：AI is exhausting workers so much, researchers have dubbed the condition 'AI brain fry'

<https://edition.cnn.com/2026/03/13/business/ai-brain-fry-nightcap>

[3] "Using too many AI tools at once can actually make you less productive and cause 'brain fry,' study finds" (Business Insider)

中文原文：Using too many AI tools at once can actually make you less productive and cause 'brain fry,' study finds

<https://www.businessinsider.com/ai-brain-fry-study-agents-uses-2026-3>

[4] "Companies Start Tracking Their Workers' AI Token Usage" (IBL News)

中文原文：Companies Start Tracking Their Workers' AI Token Usage

<https://iblnews.org/story/companies-start-tracking-their-workers-ai-token-usage>

[5] "Big tech's 'beasts of burden' are forced to use AI" (Huxiu)

中文原文：大厂“牛马”，被迫用AI

<https://www.huxiu.com/article/4848066.html>

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